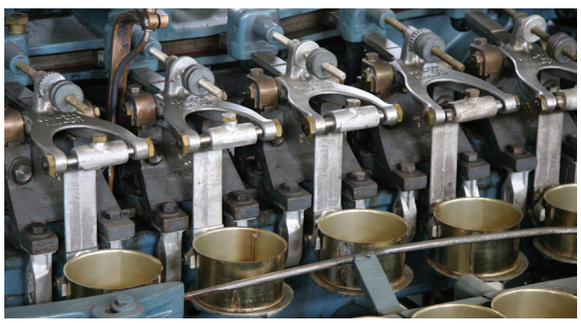
b. **Over Flow Method:** Juice is heated to a temperature about 2.5°C higher than the pasteurization temperature and filled in hot sterilized bottles. The sealed bottles are sterilized at a temperature 2.5°C lower than filling temperature and then cooled.

c. **Flash Pasteurization:** The juice is heated rapidly to a temperature of about 5.5°C higher than the pasteurization temperature and kept at this temperature for about a minute. This method has been developed specially for canning of natural orange juice, grape juice and apple juice. It has the advantages of minimizing the flavor loss and preserving the vitamins.

2. Canning: Canning is the process in which the foods are heated in hermetically sealed (airtight) jars or cans to a temperature that destroys microorganisms and inactivates enzymes that can cause food spoilage.

The general steps to be adopted for canning foods are cleaning, blanching, filling, exhausting, sealing, sterilizing, cooling and labeling.



▲ Plate 6.2 Canning

i. **Cleaning:** It is the first step in canning. Thorough cleaning of the product to be canned helps to remove most of the spoilage organisms.

ii. **Blanching:** This process serves as an additional hot water wash. It inactivates the food enzymes and fixes the natural colour of the product. It also softens the fibrous plant tissues and facilitates removal of skin. In this process the raw food material is immersed in hot water or exposed to live steam for 2–5 minutes and immediately dipped into cold water to prevent further exposure to heat.

iii. **Filling:** Either manually or by using machinery the contents can be filled in the cans. A headspace of 6–9mm depth above the level of food in the can must be left.

iv. **Exhausting:** Gases are expelled by passing the open can containing the food through an exhaust box in which hot water or steam is used. It expands the food and expels air and other gases from contents and also from the head space area of the can.

v. **Sealing:** The exhausted containers are immediately sealed to avoid re-contamination.

vi. **Sterilization:** To bring about complete sterilization, thermal process is carried out. This ensures the destruction of spoilage microorganisms. This is usually done by the application of steam under pressure.

vii. **Cooling:** The containers are cooled rapidly to check the action of heat and prevent unnecessary softening of the food or change in the colour of the contents. It can be done by means of air or water.

viii. **Labelling:** The containers are labelled with nutritional characteristics of the food inside.

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